

AISI 4320 Steel, normalized at 895°C (1640°F)

Categories: [Metal](#); [Ferrous Metal](#); [Alloy Steel](#); [AISI 4000 Series Steel](#); [Low Alloy Steel](#); [Carbon Steel](#); [Low Carbon Steel](#)

Material Notes: Heat treatment: Direct quench for pot - carburized at 925°C (1700°F) for 8 hours, quenched in agitated oil, 150°C (300°F) temper.

Key Words: UNS G43200, ASTM A322, ASTM A331, ASTM A505, ASTM A519, ASTM A535

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	7.85 g/cc	0.284 lb/in ³	

Mechanical Properties	Metric	English	Comments
Hardness, Brinell	235	235	
Hardness, Knoop	259	259	Converted from Brinell
Hardness, Rockwell B	97	97	Converted from Brinell
Hardness, Rockwell C	21	21	Converted from Brinell
Hardness, Vickers	247	247	Converted from Brinell
Tensile Strength, Ultimate	793 MPa	115000 psi	
Tensile Strength, Yield	460 MPa	66700 psi	
Elongation at Break	20.8 %	20.8 %	in 50 mm
Reduction of Area	51 %	51 %	
Modulus of Elasticity	205 GPa	29700 ksi	Typical for steel
Bulk Modulus	160 GPa	23200 ksi	Typical for steel
Poissons Ratio	0.29	0.29	Calculated
Machinability	60 %	60 %	annealed and cold drawn. Based on 100% machinability for AISI 1212 steel.
Shear Modulus	80.0 GPa	11600 ksi	Typical for steels.
Izod Impact	73.0 J	53.8 ft-lb	

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.0000245 ohm-cm	0.0000245 ohm-cm	Typical 4000 series steel

Thermal Properties	Metric	English	Comments
CTE, linear	11.3 µm/m-°C @Temperature -18.0 - 95.0 °C	6.28 µin/in-°F @Temperature -0.400 - 203 °F	condition of specimen unknown
Specific Heat Capacity	0.475 J/g-°C	0.114 BTU/lb-°F	Typical 4000 series steel
Thermal Conductivity	44.5 W/m-K	309 BTU-in/hr-ft ² -°F	Typical steel

Component Elements Properties	Metric	English	Comments
Carbon, C	0.17 - 0.22 %	0.17 - 0.22 %	
Chromium, Cr	0.40 - 0.60 %	0.40 - 0.60 %	
Iron, Fe	95.855 - 96.98 %	95.855 - 96.98 %	As remainder
Manganese, Mn	0.45 - 0.65 %	0.45 - 0.65 %	
Molybdenum, Mo	0.20 - 0.30 %	0.20 - 0.30 %	
Nickel, Ni	1.65 - 2.0 %	1.65 - 2.0 %	
Phosphorus, P	<= 0.035 %	<= 0.035 %	
Silicon, Si	0.15 - 0.30 %	0.15 - 0.30 %	
Sulfur, S	<= 0.040 %	<= 0.040 %	

[References](#) for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.